A ‘segmental boost’ for prosodic structure priming

Word stress is commonly thought of as a ‘suprasegmental’, meaning that its representation is independent of segmental content. However, priming studies often point to a representation of stress in the lexicon that is tightly integrated with specific segments and their phonetics (Protopapaps et al., 2016; Schiller et al., 2004). These studies find stress priming in lexical decision only when there is substantial segmental overlap between prime and target ([ǽdmə] facilitates *ádmiral* but not *óctopus*). This study provides evidence *for* the suprasegmental representation of stress, as well as for an explanation of the lexical decision results in a “segmental boost” similar to the lexical boost observed in syntactic structural priming (Tooley & Traxler, 2010). Moore-Cantwell et al. (2024) demonstrate production priming of stress in nonwords, even when no segmental material overlaps, and argue that stress priming, like syntactic priming, has a bigger effect on structural choices than on processing times (Pickering & Ferreira, 2008). Because they only looked at nonwords, it is possible this effect is due to nonword-specific mechanisms, such as implicit learning of stress rules. This study uses pairs of orthographically ambiguous English words (e.g. *console*, *record*) to test for production priming in real words, and follows up with a lexical decision task with those same words comparing primes which overlap segmentally on the stressed syllable, on the stressless syllable only, or not at all.
Exp1. 40 pairs of English homographs were selected as targets. Each target form corresponded to an initially-stressed noun and a finally-stressed verb. Nouns and verbs were within 1 log frequency of each other and balanced for which was more frequent. For each target, primes were selected which did not share segmental material with either homograph, and did not share letters with the spelling. 60 undergraduate participants read prime-target pairs out loud while their voice was recorded, and then used the target form in a sentence, which they typed out. These sentences were then analyzed for which syntactic category, and therefore which stress pattern, the target had been interpreted as. **Results:** Participants were more likely to interpret ambiguous targets as initially-stressed when they followed an initially-stressed prime, and as final-stressed when they followed a final-stressed prime **(1)**. A mixed-effects logistic regression showed a significant effect of prime stress on participants’ choices **(***$p<0.01$,* **2)**. Interactions did not improve model fit.
Exp2. 24 of the target items from Exp1 were used in a cross-modal priming paradigm. For each, the final-stressed version and the initial-stressed version were recorded and presented to participants in a lexical decision task. Each recording was preceded by a prime, presented visually for 250 ms and then masked by # characters. Primes either matched or mismatched the target in stress, and had either (a) no segmental overlap, (b) segmental overlap in the unstressed syllable only (conceit console, tadpole console), or (c) (conjure, console, unroll console). Reciprocal RT’s were analyzed, as these were normally distributed. **Results:** A mixed-effects regression run on reciprocal RT’s found a significant interaction between stress match and stress overlap, where stress match affected reaction times most when prime and target overlapped in their stressed syllable. Target stress also mattered: Final stressed targets showed a larger priming effect than did initial stressed targets.
Conclusion. Recently encountered stresses affect stress choices on real words, just as they do with nonwords. This effect occurs even in the absence of segmental overlap between prime and target. However, in a lexical decision task, only those with shared segmental material in the stressed syllable were facilitated. This indicates that stress is represented suprasegmentally on words just as in nonwords, but that its priming effects are small in comprehension, and boosted by segmental overlap, just as syntactic structural priming is boosted by lexical overlap.

(1) Exp. 1 number of responses

		TARGET	
		I cónsole	F consóle
PRIME	I hýbrid	576	500
	F shampóo	581	648

(2) Exp. 1 Logistic regression results

target stress ~ prime stress * prime frequency + prime stress * target bias + (1|participant) + (1|item)		
	Estimate	p value
prime stress	**1.11**	**0.024**
prime frequency	0.30	0.18
target bias	0.74	0.25
prime stress * prime frequency	-0.40	0.12
prime stress * target bias	**0.52**	**0.01**

(3) Exp. 2 Raw reaction time means by condition

(a) No overlap

		TARGET	
		I cónsole	F consóle
PRIME	I hýbrid	1353	1411
	F shampóo	1368	1399

(b) Unstressed overlap

		TARGET	
		I cónsole	F consóle
	I tádpole	1402	1419
	F concéit	1368	1390

(c) Stressed overlap

		TARGET	
		I cónsole	F consóle
	I cónjure	1392	**1423**
	F unróll	1356	**1373**

(4) Exp. 2 regression (significant effects only)		
*1/RT ~ stress match * target stress * prime frequency * target frequency + (1|participant) + (1|item)*		
	Estimate	**t value**
prime frequency	$-2.37e^{-04}$	-2.00
stress match * target frequency	$-3.29e^{-04}$	-2.21
stress match * target stress	$1.00e^{-04}$	-1.98
stress match * overlap (stressed syllable)	$-8.37e^{-04}$	-2.19
target frequency * prime frequency	$9.10e^{-05}$	1.97
stress match * target frequency * target stress	$4.48e^{-04}$	2.00
stress match * overlap (stressed syllable) * target frequency	$4.39e^{-04}$	2.65
stress match * overlap (stressed syllable) * target stress	$1.47e^{-04}$	2.58
stress match * overlap (stressed syllable) * prime frequency	$5.37e^{-04}$	2.03
stress match * overlap (str) * target stress * target frequency	$-6.97e^{-04}$	-2.81
stress match * overlap (unstr) * target stress * target frequency	$-5.57e^{-04}$	-2.12
stress match * overlap (str) * prime frequency * target frequency	$-2.83e^{-04}$	-2.42
stress match * overlap (str) * target stress * prime frequency	$-1.04e^{-04}$	-2.65
stress match * overlap (str) * target stress * prime frequency * target frequency	$4.70e^{-04}$	2.82

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